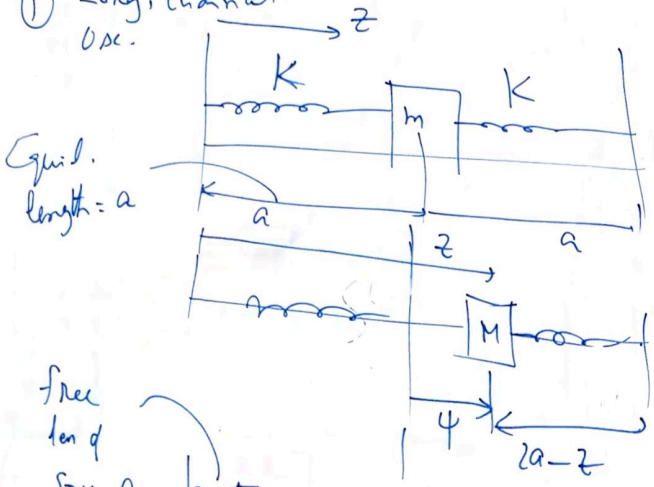
25.7.16

Waves Lec 2 (P1)

Berkeley Phys. Vol. 3 Waves

Two-springs-Mass Sys

① Longitudinal Osc.



Inequil.

$T_0 = K(a - a_0)$

Assum $a_0 < a$

(P2)

Pos. of M

$z = a + \psi$

ψ = displacement

Vel: $\dot{z} = \dot{\psi}$

acc = $\ddot{z} = \ddot{\psi}$

Tension in string 1

$T_1 = K(z - a_0)$

$= K(a + \psi - a_0)$

Restoring force from string 1

$F_1 = -T_1$

String 2 $T_2 = K(2a - z - a_0)$

$= K(a - \psi - a_0)$

(P3)

restoring force from Sp 2

$F_2 = T_2$

E.o.M:

$M\ddot{\psi} = F_1 + F_2$

$= -T_1 + T_2$

$= -2K\psi$

$\ddot{\psi} = -\frac{2K}{M}\psi = -\omega^2\psi$

$\omega^2_{Long} = \frac{2K}{M}$

* Independent of a_0,

$a - |\psi_{max}| > a_0$

OR, $|\psi_{max}| < a - a_0$

This true for slinkies for which min. length is a_0

A slinky cannot be compressed to a length less than a_0 with the spring const K.

(P7)

$\omega_L^2 = \frac{2K}{M}$

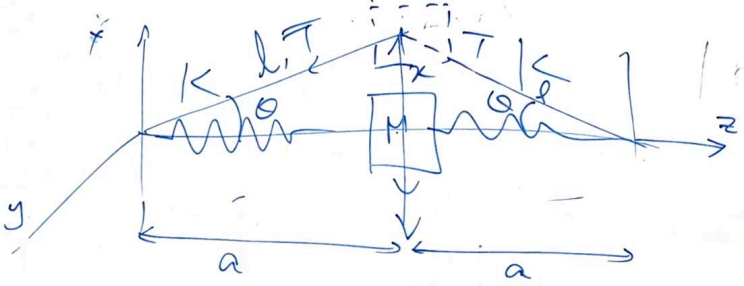
$\omega_T^2 = \frac{2K}{M}(1 - \frac{a_0}{a})$

$\frac{\omega_L}{\omega_T} = \frac{1}{(1 - a_0/a)^{1/2}}$

> 1

Transverse oscillations

(P4)



Let M get displaced along x-direction, by x

$$l = \sqrt{a^2 + x^2}, \quad \sin\theta = x/l$$

String 1: $T_1 = k(l - a_0)$,

Restoring force $F_1 = -T_1 \sin\theta = -k(l-a_0)\frac{x}{l}$

$$= -kx\left(1 - \frac{a_0}{l}\right)$$

String 2: $T_2 = k(l - a_0)$ (P5)

Restoring force $F_2 = -T_2 \sin\theta$

$$= -kx\left(1 - \frac{a_0}{l}\right)$$

Net restoring force

$$F = F_1 + F_2 = -2kx\left(1 - \frac{a_0}{l}\right)$$

EoM:

$$M\frac{d^2x}{dt^2} = -2kx\left(1 - \frac{a_0}{\sqrt{a^2+x^2}}\right)$$

1 Slinky approximation: $a_0 \ll a$

$$\Rightarrow a_0 \approx 0$$

$$M\ddot{x} = -2kx \quad \Big| \quad \omega^2 = \frac{2k}{M}$$

$$\ddot{x} = -\frac{2k}{M}x$$

(P6)

Approx 2: Small osc. Approx

$$l = \sqrt{a^2 + x^2} = a\left[1 + \frac{x^2}{a^2}\right]^{1/2}$$

For small osc. $|x| \ll a$, or

$$\epsilon \equiv |x|/a \ll 1$$

$$\frac{a_0}{l} = \frac{a_0}{a}(1+\epsilon)^{-1/2}$$

$$= \frac{a_0}{a}\left(1 - \frac{1}{2}\frac{x^2}{a^2} + \dots\right)$$

$$M\ddot{x} = -2kx\left[1 - \frac{a_0}{a}\left(1 - \frac{1}{2}\frac{x^2}{a^2} + \dots\right)\right]$$

$$\approx -2k\left(1 - \frac{a_0}{a}\right)x$$

$$= -2\frac{\overbrace{k(a-a_0)}^{T_0}}{a}x \qquad (l \approx a)$$

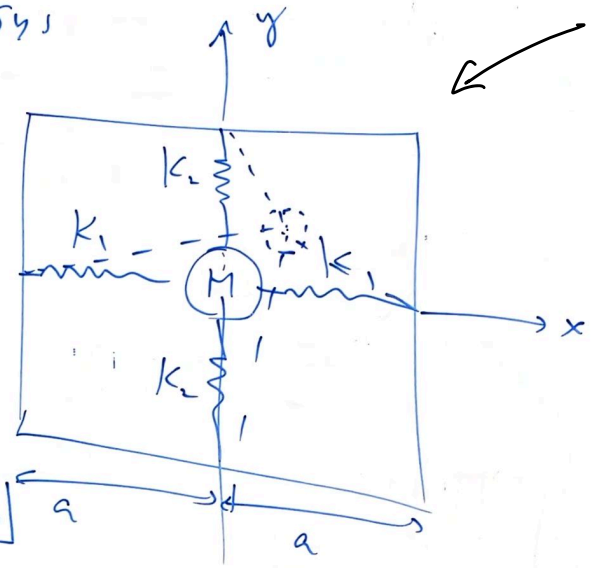
$$M\ddot{x} = -\frac{2T_0}{a}x$$

$$\ddot{x} = -\frac{2T_0}{Ma}x \Rightarrow \omega_{Tr}^2 = \frac{2T_0}{Ma} = \frac{2k}{M}(1 - a_0/a)$$

25.7.26

Warm Lec 20

Simplest sys with two degrees of freedom



For small x, y

x-mode osc.

(1) $M\ddot{x} \approx -2K_1 x \Rightarrow \omega_1^2 = \frac{2K_1}{M}$

(2) $M\ddot{y} \approx -2K_2 y \Rightarrow \omega_2^2 = \frac{2K_2}{M}$

y-mode osc.

} Two modes

Eqs (1) and (2) do not have terms that couple x and y motions. They are said to be decoupled and can be solved independently.

In this example, it is assumed that $a = a_0$. In this case, when the mass M is moved along x direction, restoring force from K_2 spring are:

$F_x(K_2\text{-springs}) \approx -2K_2 x^3$

(please see the remarks in the example on transverse oscillation)

The non-linear term may be ignored in the linear approx.

$\therefore$ F_x comes only from K_1 springs:

$$F_x = -\frac{2K_1 \cdot x}{M}$$

A similar arguments holds for F_y

↓

Their solutions are:

$$x = A_1 \cos(\omega_1 t + \varphi_1)$$

$$y = A_2 \cos(\omega_2 t + \varphi_2)$$